Numerical Methods in Macroeconomics

Cardiff Lecture, Part 1

Alessandro Di Nola

University of Birmingham

Outline

- The stochastic growth model: quick review
- Equilibrium conditions and deterministic steady state
- Approximation with first-order perturbation
- Solution by undetermined coefficients
- Simulation and impulse responses
- MATLAB implementation

The Stochastic Growth Model

- We solve the stochastic neoclassical growth model using the first-order perturbation method.
- The workflow is:
  1. Write down the equilibrium conditions.
  2. Compute the deterministic steady state.
  3. Take a linear or log-linear approximation around the deterministic steady state.
  4. Solve the linear rational expectations system with the method of undetermined coefficients (main reference: Uhlig, 1999)
  5. Simulate the model and compute impulse responses.
- We will see later how to automate these steps with Dynare.

Step 1 - The Planner's Problem

- The planner chooses the sequence $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ to solve

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \quad E_0 \left\{ \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma} \right\}$$

  subject to

$$c_t + k_{t+1} \leq z_t k_t^{\alpha} + (1-\delta) k_t, \qquad t = 0, 1, 2, \ldots$$

- c_t is consumption, k_t is beginning-of-period capital, and z_t is productivity.
- Parameters: $\beta \in (0,1)$, $\sigma > 0$, $\alpha \in (0,1)$, $\delta \in (0,1]$.

Step 1 - The Planner's Problem

- The planner chooses the sequence $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ to solve

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \quad E_0 \left\{ \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma} \right\}$$

  subject to

$$c_t + k_{t+1} \leq z_t k_t^{\alpha} + (1-\delta) k_t, \qquad t = 0, 1, 2, \ldots$$

- c_t is consumption, k_t is beginning-of-period capital, and z_t is productivity.
- Parameters: $\beta \in (0,1)$, $\sigma > 0$, $\alpha \in (0,1)$, $\delta \in (0,1]$.
- Q: Why do we consider the planner's problem if we are ultimately interested in a competitive equilibrium?

Step 1 - Technology and First-Order Conditions

- Productivity follows

$$\log z_{t+1} = \rho \log z_t + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \text{i.i.d. } N(0, \sigma_\varepsilon^2)$$

  with $\rho \in (0, 1)$.

- The first-order conditions w.r.t. c_t and k_{t+1} are

$$c_t^{-\sigma} = \lambda_t$$

$$\lambda_t = \beta E_t \left[\lambda_{t+1} \left(\alpha z_{t+1} k_{t+1}^{\alpha-1} + 1 - \delta \right) \right]$$

  where λ_t is the Lagrange multiplier on the resource constraint. Derivation

- These conditions hold together with the resource constraint.

Step 1 - Equilibrium Conditions

- There are three endogenous variables:

$$c_t, \qquad k_t, \qquad z_t.$$

- Therefore we must have three equilibrium equations.
- Among the endogenous variables, k_t and z_t are state variables, while c_t is the control variable.
- ε_t is an exogenous shock.
- The equilibrium system is

$$c_t^{-\sigma} = \beta E_t \left[c_{t+1}^{-\sigma} \left(\alpha z_{t+1} k_{t+1}^{\alpha-1} + 1 - \delta \right) \right] \tag{EE}$$

$$c_t + k_{t+1} = z_t k_t^{\alpha} + (1-\delta) k_t \tag{RC}$$

$$\log z_t = \rho \log z_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \text{i.i.d. } N(0, \sigma_\varepsilon^2) \tag{Shock}$$

Step 2 - Deterministic Steady State

- In the deterministic steady state, $\varepsilon_t = 0$ and

$$c_t = \bar{c}, \qquad k_t = \bar{k}, \qquad z_t = \bar{z}$$

  for all t.

- The steady-state system is

$$1 = \beta \left(\alpha \bar{z} \bar{k}^{\alpha - 1} + 1 - \delta \right) \tag{EE-SS}$$

$$\bar{c} + \bar{k} = \bar{z} \bar{k}^{\alpha} + (1 - \delta) \bar{k} \tag{RC-SS}$$

$$\log \bar{z} = \rho \log \bar{z} \tag{Shock-SS}$$

- Since $\rho \in (0, 1)$, we have $\bar{z} = 1$.

Step 3 - Steady-State Capital and Consumption

- Solving (EE-SS) for steady-state capital,

$$\bar{k} = \left(\frac{\alpha \bar{z}}{\frac{1}{\beta} - 1 + \delta} \right)^{\frac{1}{1-\alpha}} \tag{1}$$

- Steady-state consumption follows from (RC-SS)

$$\bar{c} = \bar{z}\bar{k}^{\alpha} - \delta \bar{k} \tag{2}$$

- Steady-state output is

$$\bar{y} = \bar{z}\bar{k}^{\alpha} \tag{3}$$

Step 3 - First-Order Approximation

- If $f(x_t, y_t) = 0$, a first-order Taylor approximation around $(\bar{x}, \bar{y})$ is

$$f(x_t, y_t) \approx f(\bar{x}, \bar{y}) + f_x(\bar{x}, \bar{y})(x_t - \bar{x}) + f_y(\bar{x}, \bar{y})(y_t - \bar{y})$$

- Define log deviations as

$$\hat{x}_t \equiv \log\left(\frac{x_t}{\bar{x}}\right) \approx \frac{x_t - \bar{x}}{\bar{x}}$$

- A useful rule is

$$x_t = \bar{x}e^{\hat{x}_t} \approx \bar{x}(1 + \hat{x}_t)$$

Step 3 - Linearized Resource Constraint

- Define

$$\hat{c}_t = \log\left(\frac{c_t}{\bar{c}}\right), \qquad \hat{k}_t = \log\left(\frac{k_t}{\bar{k}}\right), \qquad \hat{z}_t = \log\left(\frac{z_t}{\bar{z}}\right)$$

- The log-linearized resource constraint is

$$\frac{\bar{c}}{\bar{k}}\hat{c}_t + \hat{k}_{t+1} - \frac{1}{\beta}\hat{k}_t - \frac{1}{\alpha}\left(\frac{1}{\beta} - 1 + \delta\right)\hat{z}_t = 0 \qquad \text{(RC-LIN)}$$

- Here

$$\frac{\bar{y}}{\bar{k}} = \bar{z}\bar{k}^{\alpha-1} = \frac{1}{\alpha}\left(\frac{1}{\beta} - 1 + \delta\right)$$

Step 3 - Linearized Euler Equation

- Define the gross return on capital and its log deviation:

$$R_{t+1} \equiv \alpha z_{t+1} k_{t+1}^{\alpha-1} + 1 - \delta \qquad \hat{R}_{t+1} \equiv \frac{R_{t+1} - \bar{R}}{\bar{R}}$$

- Since $\bar{R} = 1/\beta$, a first-order approximation of the Euler equation implies

$$\hat{c}_t = E_t \hat{c}_{t+1} - \frac{1}{\sigma} E_t \hat{R}_{t+1} \tag{4}$$

- Also,

$$\hat{R}_{t+1} = [1 - \beta(1-\delta)] \left(\hat{z}_{t+1} - (1-\alpha) \hat{k}_{t+1} \right) \tag{5}$$

- Using $E_t \hat{z}_{t+1} = \rho \hat{z}_t$, the linearized Euler equation becomes

$$\hat{c}_t = E_t \hat{c}_{t+1} - \frac{1}{\sigma} [1 - \beta(1-\delta)] \rho \hat{z}_t + \frac{1}{\sigma}(1-\alpha) [1 - \beta(1-\delta)] \hat{k}_{t+1} \tag{EE-LIN}$$

Step 3 - The Linear System

- The linearized model is given by (RC-LIN)-(EE-LIN):

$$\alpha_1 \hat{c}_t + \alpha_2 \hat{k}_{t+1} + \alpha_3 \hat{k}_t + \alpha_4 \hat{z}_t = 0$$

$$\beta_1 \hat{c}_t + \beta_2 E_t \hat{c}_{t+1} + \beta_3 \hat{z}_t + \beta_4 \hat{k}_{t+1} = 0$$

- where

$$\alpha_1 = \frac{\bar{c}}{\bar{k}}, \quad \alpha_2 = 1,$$
$$\alpha_3 = -\frac{1}{\beta}, \quad \alpha_4 = -\frac{1}{\alpha}\left(\frac{1}{\beta} - 1 + \delta\right)$$

$$\beta_1 = 1, \quad \beta_2 = -1,$$
$$\beta_3 = \frac{1}{\sigma}\left[1 - \beta(1-\delta)\right]\rho, \quad \beta_4 = -\frac{1}{\sigma}(1-\alpha)\left[1 - \beta(1-\delta)\right]$$

Step 4 - Policy Functions

- Guess linear decision (policy) rules:

$$\hat{c}_t = \eta_{ck}\hat{k}_t + \eta_{cz}\hat{z}_t \qquad \text{(POL-C)}$$

$$\hat{k}_{t+1} = \eta_{kk}\hat{k}_t + \eta_{kz}\hat{z}_t \qquad \text{(POL-K)}$$

- Taking expectations,

$$E_t\hat{c}_{t+1} = \eta_{ck}\eta_{kk}\hat{k}_t + \left(\eta_{ck}\eta_{kz} + \eta_{cz}\rho\right)\hat{z}_t$$

Procedure:

- Substitute (POL-C)-(POL-K) into the two linearized equilibrium equations.
- Resulting expressions must hold for every possible value of the state variables $\hat{k}_t$ and $\hat{z}_t$.
- Therefore, in each equation, the coefficient on $\hat{k}_t$ must be zero and the coefficient on $\hat{z}_t$ must also be zero.
- This gives four equations for the four unknown coefficients

Step 4 - Undetermined Coefficients

$$\alpha_1 \eta_{ck} + \alpha_2 \eta_{kk} + \alpha_3 = 0 \tag{UC-1}$$

$$\alpha_1 \eta_{cz} + \alpha_2 \eta_{kz} + \alpha_4 = 0 \tag{UC-2}$$

$$\beta_1 \eta_{ck} + \beta_2 \eta_{ck} \eta_{kk} + \beta_4 \eta_{kk} = 0 \tag{UC-3}$$

$$\beta_1 \eta_{cz} + \beta_2 \left(\eta_{ck} \eta_{kz} + \eta_{cz} \rho \right) + \beta_3 + \beta_4 \eta_{kz} = 0 \tag{UC-4}$$

- From equation (UC-1) we have

$$\eta_{ck} = -\frac{\alpha_2 \eta_{kk} + \alpha_3}{\alpha_1} \tag{eta-ck}$$

- Substituting (eta-ck) into equation (UC-3), we obtain the quadratic equation for η_{kk}:

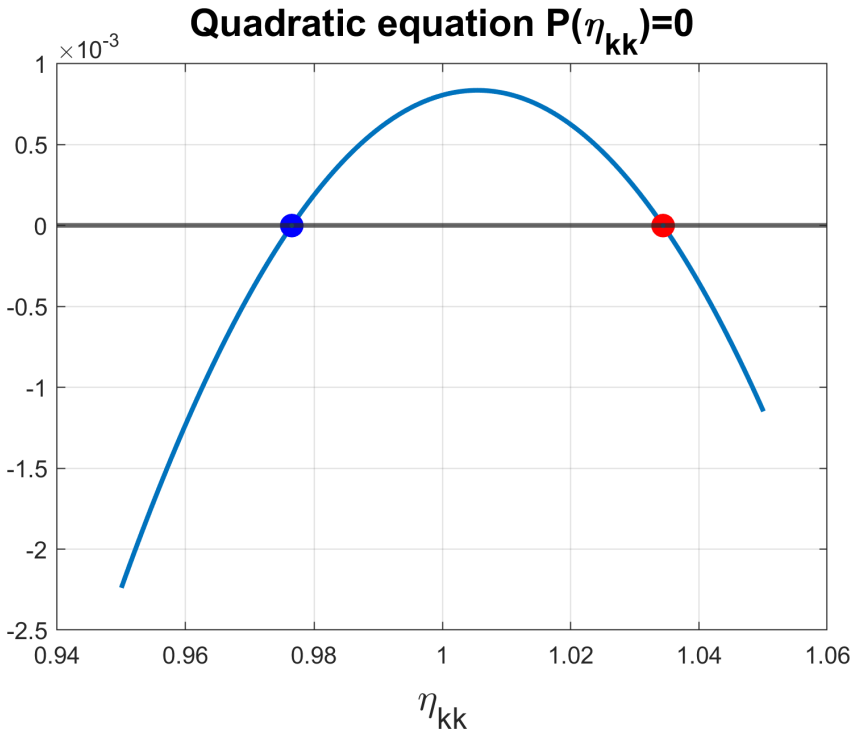
$$\beta_2 \alpha_2 \eta_{kk}^2 + \left(\beta_1 \alpha_2 + \beta_2 \alpha_3 - \beta_4 \alpha_1 \right) \eta_{kk} + \beta_1 \alpha_3 = 0$$

Step 4 - Stable and Unstable Root

- Quadratic equation for η_{kk}:

$$P(\eta_{kk}) \equiv \beta_2\alpha_2\eta_{kk}^2 + (\beta_1\alpha_2 + \beta_2\alpha_3 - \beta_4\alpha_1)\,\eta_{kk} + \beta_1\alpha_3 = 0$$

- It can be shown that this equation has two roots: one root with modulus less than 1 (stable) and the other with modulus greater than 1 (unstable).



Step 4 - Closed-Form Coefficients

- Select the stable root of the quadratic, the one satisfying $|\eta_{kk}| < 1$.
- Then recover the remaining coefficients:

$$\eta_{cz} = -\frac{\alpha_2 \eta_{kz} + \alpha_4}{\alpha_1}$$

$$\eta_{kz} = \frac{(\beta_1 + \rho\beta_2)\,\alpha_4 - \alpha_1\beta_3}{\alpha_1\beta_4 - \beta_1\alpha_2 + \beta_2\,(\alpha_1\eta_{ck} - \alpha_2\rho)}$$

- Once we have the four coefficients, we can simulate the model and compute impulse responses.

Step 5 - Simulation and Impulse Responses

- Start from $\hat{z}_0 = 0$ and $\hat{k}_0 = 0$.
- For simulations, generate shocks $\{\varepsilon_t\}_{t=1}^{T}$ from $N(0, \sigma_\varepsilon^2)$ and use

$$\hat{z}_t = \rho \hat{z}_{t-1} + \varepsilon_t$$

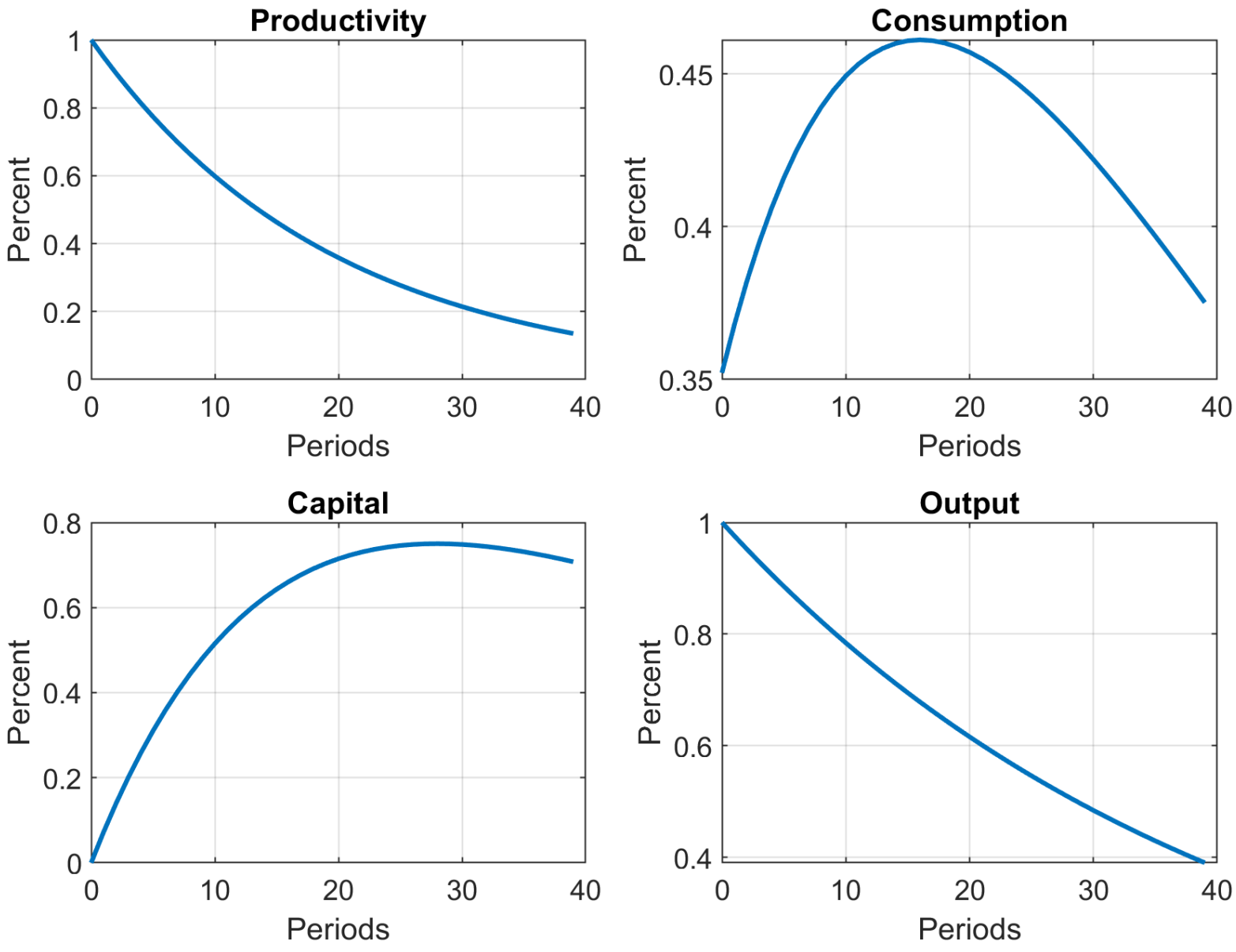
- Iterate on the policy rules:

$$\hat{c}_t = \eta_{ck} \hat{k}_t + \eta_{cz} \hat{z}_t, \qquad \hat{k}_{t+1} = \eta_{kk} \hat{k}_t + \eta_{kz} \hat{z}_t$$

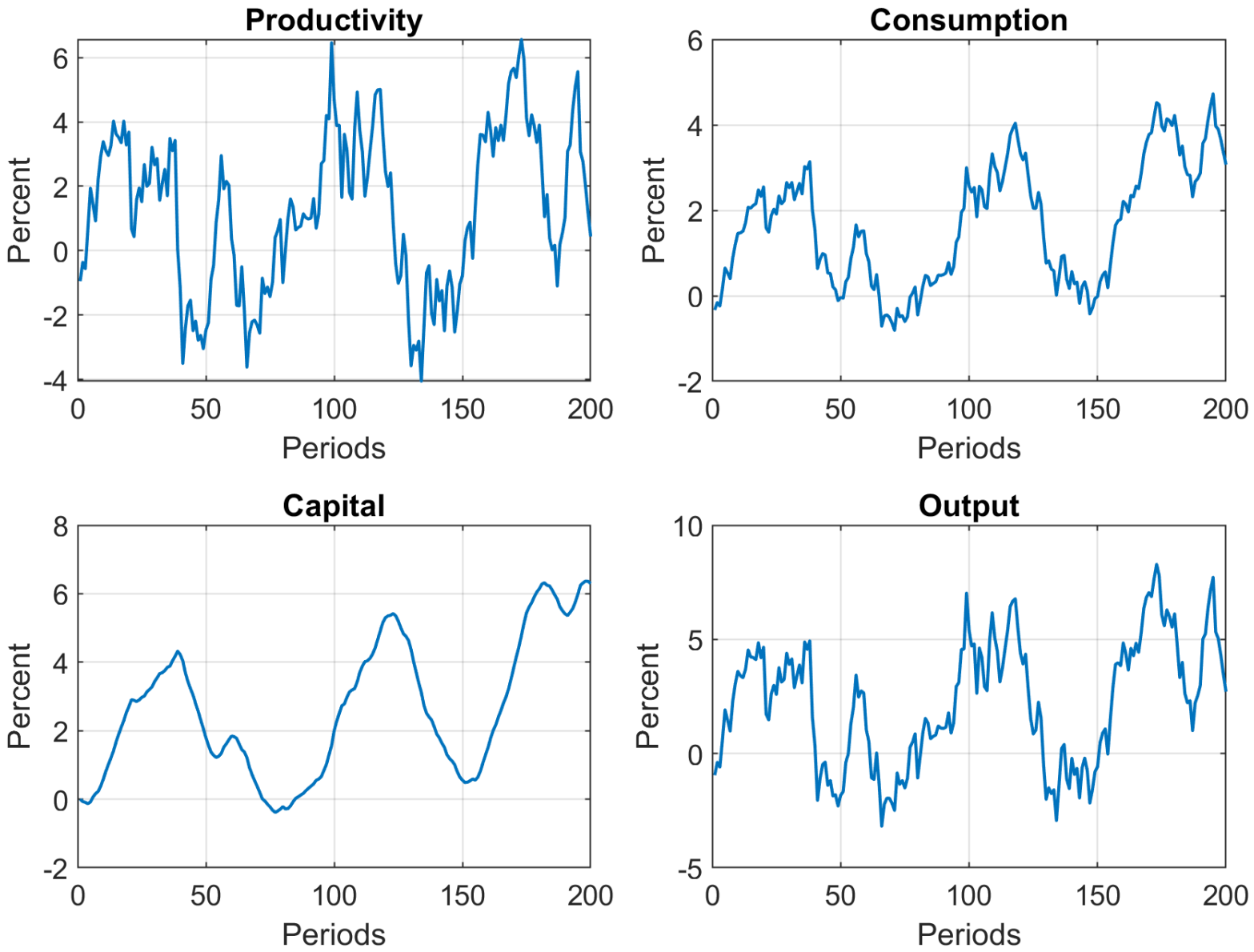
- Recover output from

$$\hat{y}_t = \hat{z}_t + \alpha \hat{k}_t$$

Impulse Response from MATLAB



Stochastic Simulation from MATLAB



MATLAB Implementation

- The teaching script is `codes/main_undet_coeffs.m`.
- It follows exactly the structure of the notes:
  1. set parameters,
  2. compute the steady state,
  3. build the linear coefficients,
  4. solve for $\eta_{ck}, \eta_{cz}, \eta_{kk}, \eta_{kz}$,
  5. produce an IRF and a stochastic simulation.
- The auxiliary function `codes/fun_ss.m` computes

$$\bar{k}, \quad \bar{c}, \quad \bar{z}, \quad \bar{y}.$$

Appendix: Euler Equation Derivation

- Consider the planner's problem:

$$\max_{\{c_t, k_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}$$

subject to

$$c_t + k_{t+1} = z_t k_t^{\alpha} + (1-\delta) k_t.$$

- Attach multiplier λ_t to the resource constraint. The intertemporal Lagrangian is

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\sigma}}{1-\sigma} + \lambda_t \left(z_t k_t^{\alpha} + (1-\delta) k_t - c_t - k_{t+1} \right) \right].$$

Appendix: Euler Equation Derivation

- FOC with respect to c_t:

$$\frac{\partial \mathcal{L}}{\partial c_t}: \quad c_t^{-\sigma} - \lambda_t = 0 \quad \Rightarrow \quad \lambda_t = c_t^{-\sigma}.$$

- FOC with respect to k_{t+1}:

$$\frac{\partial \mathcal{L}}{\partial k_{t+1}}: \quad -\lambda_t + \beta E_t \left[\lambda_{t+1} \left(\alpha z_{t+1} k_{t+1}^{\alpha-1} + 1 - \delta\right)\right] = 0.$$

- Substitute out the multiplier using $\lambda_t = c_t^{-\sigma}$ and $\lambda_{t+1} = c_{t+1}^{-\sigma}$:

$$c_t^{-\sigma} = \beta E_t \left[c_{t+1}^{-\sigma} \left(\alpha z_{t+1} k_{t+1}^{\alpha-1} + 1 - \delta\right)\right].$$

- This is the Euler equation in the main text. back

References

- Uhlig, H. (1999). "A Toolkit for Analyzing Nonlinear Dynamic Stochastic Models Easily." In R. Marimon and A. Scott (eds.), *Computational Methods for the Study of Dynamic Economies*, Oxford University Press, 30–61.
- Miao, J. (2020). *Economic Dynamics in Discrete Time*, 2nd ed. Cambridge, MA: The MIT Press.